

1. Which transformations listed are required to obtain the graph of $y = x^2 + \frac{1}{2}x - 3$ from the graph of $y = 4x^2 + x$ 1
- A. Horizontal dilation by a factor of 2; vertical translation of 3 units upwards
- B. Horizontal dilation by a factor of 2; vertical translation of 3 units downwards**
- C. Horizontal dilation by a factor of $\frac{1}{2}$; vertical translation of 3 units upwards
- D. Horizontal dilation by a factor of $\frac{1}{2}$; vertical translation of 3 units downwards

$$y = 4x^2 + x \xrightarrow{x \rightarrow \frac{x}{2}} y = 4\left(\frac{x}{2}\right)^2 + \frac{x}{2} \xrightarrow{y \rightarrow y+3} y+3 = x^2 + \frac{1}{2}x$$

$$x^2 + \frac{1}{2}x \quad y = x^2 + \frac{1}{2}x - 3$$

2. The point $A(4, 21)$ lies on the graph of $y = g(x)$. 3
- It is known that the graph of $y = g(x)$ is obtained from transforming the graph of $y = f(x)$ such that $g(x) = -2f(4x+4)+1$.
- Find the coordinates of the point on the graph of $y = f(x)$ which is mapped to the point A .

To get from $f(x) \rightarrow g(x)$

H.D. 4

H.T. Left 1

V.D. 2

V.T. up 1

Reverse for $(4, 21)$

Down 1

(4, 20)

V.D. $\frac{1}{2}$

(4, 10)

Right 1

(5, 10)

H.D. 4

(20, 10)

3. The point $P(-2, 3)$ lies on $y = f(x)$. What are the coordinates of the image point, P' on the graph of $y = 3f(x-4)+1$ 1

$$x' = -2 + 4$$

$$= 2$$

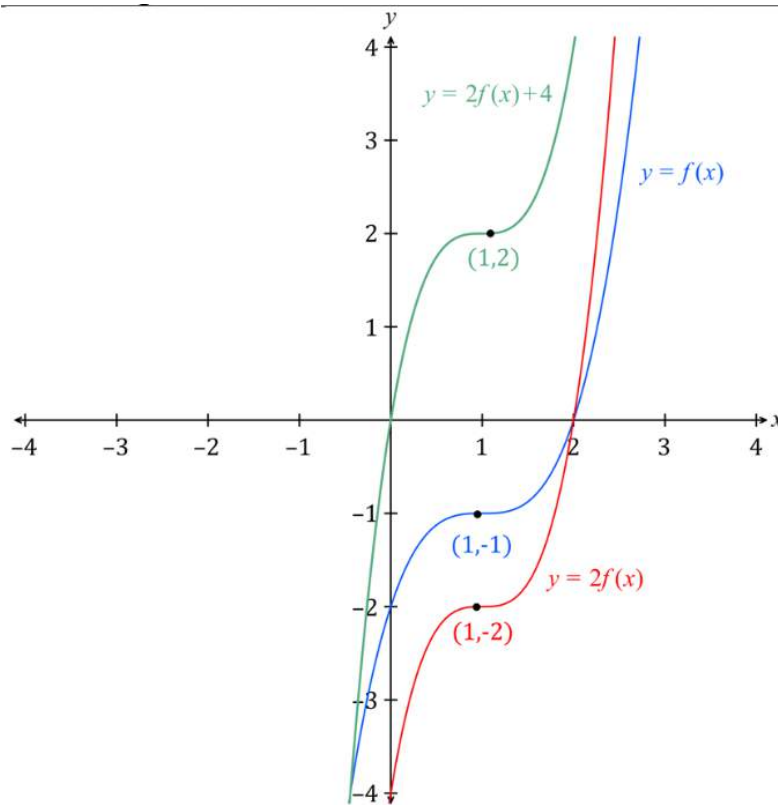
$$y' = 3 \times 3 + 1$$

$$= 10$$

4. Consider the graph of $y = f(x)$ as shown.
Draw sketches of the following functions on the above number plane. Clearly label each sketch. Indicate the x -intercepts and coordinates of the point of inflexion.

- (a) $y = 2f(x)$
(b) $y = 2f(x) + 4$

2
2



5. Describe three transformations which, when applied in succession, change the graph of $y = x^2 + 5$ to the graph with equation $y = \left(\frac{x+1}{2}\right)^2$.

$$y = x^2 + 5 \rightarrow x^2 \rightarrow \left(\frac{x}{2}\right)^2 \rightarrow \left(\frac{x+1}{2}\right)^2$$

- 5 units down (1)
- H.Dilation by factor of 2 (1)
- 1 unit left (1)

OR: $y = \left(\frac{x}{2} + \frac{1}{2}\right)^2$

- 5↓, $\leftarrow \frac{1}{2}$, 2x H.D.

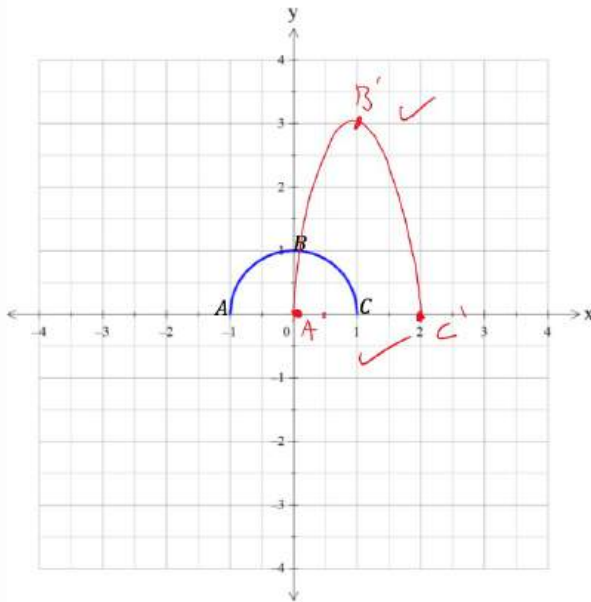
OR: • ↓ 5, $\leftarrow 1$, D× $\frac{1}{4}$ vertically

OR: • ↓ 5, D× $\frac{1}{4}$, $\leftarrow 1$

6. The function $y = f(x)$ is graphed below. On the same set of axes, sketch the transformed function clearly showing the transformed points of A, B and C .

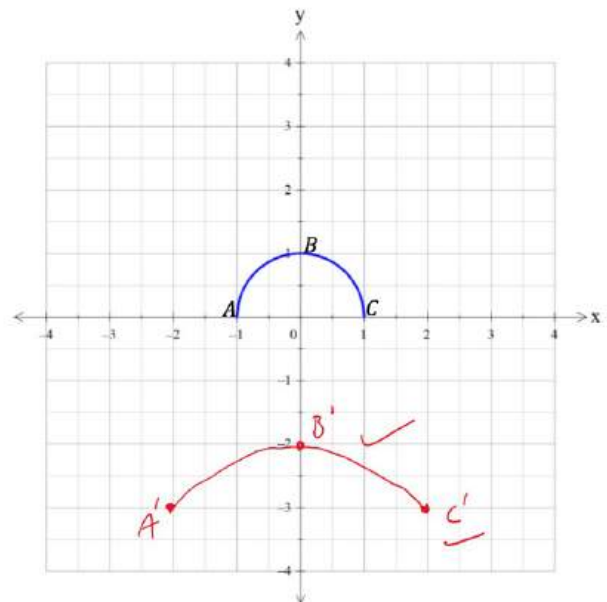
(a) $y = 3f(x - 1)$

2



(b) $y = f\left(\frac{1}{2}x\right) - 3$

2



7. P lies on the graph of $y = f(x)$ as shown on the diagram below. A transformation maps the graph of $f(x)$ to $g(x)$ such that $g(x) = 2f(1 - 2x)$. The same transformation maps the point P to P' . What are the coordinates of P' ?

1

A. $(2, -6)$

B. $\left(-\frac{1}{2}, -4\right)$

C. $\left(\frac{1}{2}, -4\right)$

D. $(-2, -4)$

$g(x) = 2f(1 - 2x)$

$= 2f\left(-2\left(x - \frac{1}{2}\right)\right)$

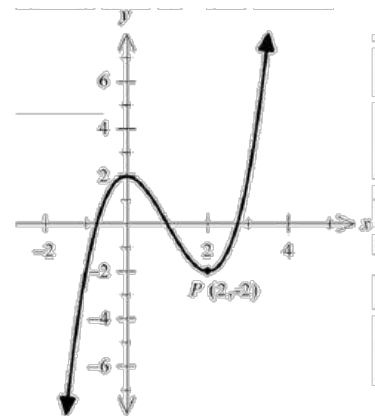
$f(-2x)$ stretches horizontally by a factor of $\frac{1}{2}$ and reflects in the y -axis
ie $(2, -2)$ maps to $(-1, -2)$

$f\left(-2\left(x - \frac{1}{2}\right)\right)$ will then translate this function $\frac{1}{2}$ unit to the right.

ie the point is now mapped to $\left(-\frac{1}{2}, -2\right)$

$2f\left(-2\left(x - \frac{1}{2}\right)\right)$ stretches vertically by a factor of 2

ie the point now maps to $\left(-\frac{1}{2}, -4\right)$



8. The function $y = \log_e x$ is transformed to $y = \log_e(3x - 3)$.

1

Which of the following describes the transformation that took place?

A. Horizontal dilation factor 3, followed by translation right 1 unit.

B. Horizontal dilation factor 3, followed by translation right 3 units.

C. Horizontal dilation factor $\frac{1}{3}$, followed by translation right 1 unit.

D. Horizontal dilation factor $\frac{1}{3}$, followed by translation right 3 units.

$y = \log_e x \rightarrow y = \log_e(3x)$

Horizontal dilation factor is $\frac{1}{3}$

$y = \log_e(3x) \rightarrow y = \log_e(3x - 3)$
 $= \log_e(3(x - 1))$

Translation right 1 unit

9. The graph of $y = x^2$ is transformed into the graph of $y = 3x^2 + 24x + 33$ by the transformations:

2

- A vertical stretch with scale factor k followed by
- A horizontal translation of p units followed by
- A vertical translation of q units

Write down the values of k , p and q .

$$y = 3x^2 + 24x + 33$$

$$y = 3(x^2 + 8x) + 33$$

$$y = 3(x^2 + 8x + 16) + 33 - 3 \times 16$$

$$y = 3(x + 4)^2 - 15$$

$$\therefore k = 3, p = -4, q = -15$$

10. A graph, $g(x)$, is obtained by transforming $f(x)$ such that $g(x) = -\frac{1}{2}f(x-1) + 2$.
The point $(3, 4)$ is on the graph of $f(x)$. Where does this point correspond to on $g(x)$?

1

A. $(4, 0)$

B. $(2, 0)$

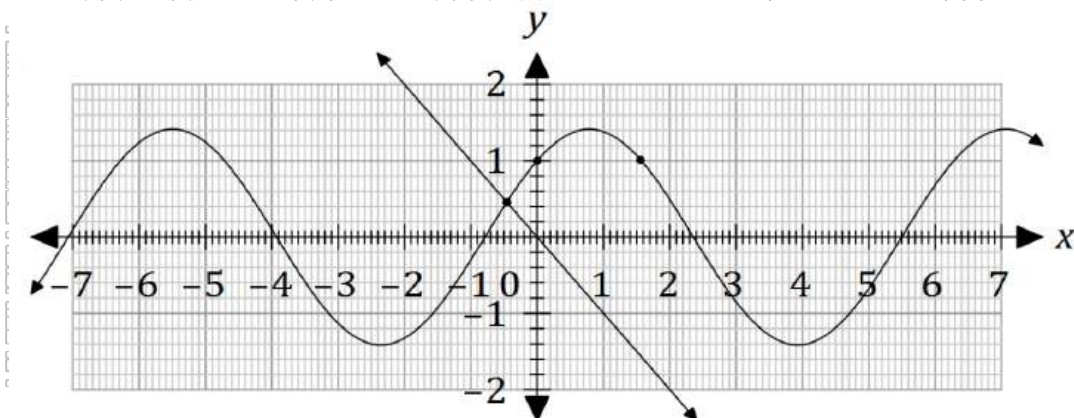
C. $(-2, 6)$

D. $(-1, 6)$

$g(x) = -\frac{1}{2}f(x-1) + 2$, hence $f(x)$ has been translated one unit to the right, dilated vertically by a factor of $\frac{1}{2}$ and reflected across the x -axis, and then translated vertically up 1.

Thus, the x -coordinate will become $3 + 1 = 4$ and the y -coordinate will become $4 \times -\frac{1}{2} + 2 = 0$. Hence $(3, 4) \rightarrow (4, 0)$ (Option A)

11. The graph of $y = \sin(x) + \cos(x)$ is shown on the axes below.



- (a) Mark the approximate solutions to the equation $\sin(x) + \cos(x) = 1$ on the domain $-\pi \leq x \leq \pi$. Write the answer(s) below, approximating your answer(s) to one decimal place.

1

$$x \approx 0, 1.6$$

- (b) Find the approximate solution(s) to $\sin(x) + \cos(x) + x = 0$ on the domain $-\pi \leq x \leq \pi$. Give your answer to the nearest tenth.

2

$$x \approx -0.5$$

12. The circle $x^2 + y^2 - 6x + 8y - 11 = 0$ is transformed by a horizontal translation to the left by 4 units and a vertical translation up 3 units. What is the centre and radius of the new circle? 3

$$x^2 - 6x + y^2 + 8y - 11 = 0$$

$$x^2 - 6x + 9 + y^2 + 8y + 16 = 11 + 9 + 16$$

$$(x-3)^2 + (y+4)^2 = 36 \quad \text{--- (1)}$$

centre $(3, -4)$

New centre
 $= (-1, -1)$

radius
 $r = 6$

13. The function $y = f(x)$ is transformed to $y = f\left(3\left(x + \frac{1}{3}\right)\right)$. Which of the following describes the transformations that took place? 1

- (A) Horizontal dilation factor of 3, followed by translation left 1 unit.
(B) Horizontal dilation factor of 3, followed by translation left $\frac{1}{3}$ unit.
(C) Horizontal dilation factor of $\frac{1}{3}$, followed by translation left 1 unit.
(D) Horizontal dilation factor of $\frac{1}{3}$, followed by translation left $\frac{1}{3}$ unit.

14. Consider the graph below of $y = \frac{k}{x+b} + c$, with y intercept at $(0, \frac{10}{3})$. Which of the following is true? 1

(A) $k < b < c$

(B) $b < k < c$

(C) $k < c < b$

(D) $c < k < b$

$$y = \frac{k}{x-3} + 2 \quad \text{sub } (0, \frac{10}{3})$$

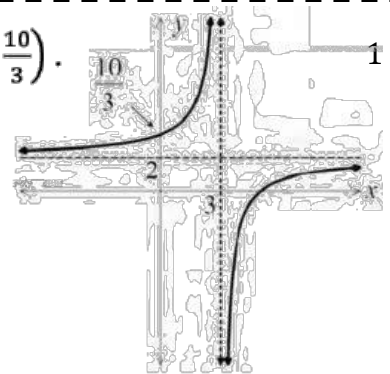
$$\frac{10}{3} = \frac{k}{-3} + 2$$

$$\frac{4}{3} = \frac{k}{-3}$$

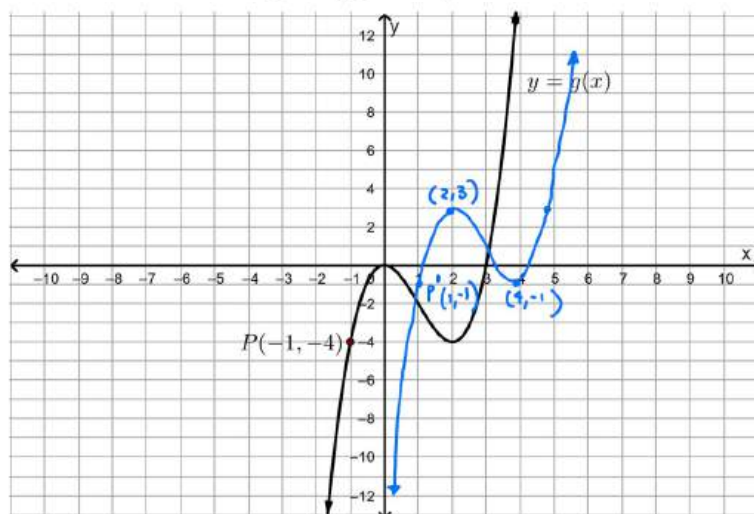
$$k = -4$$

$$b = -3$$

$$c = 2$$



15. The function $y = g(x)$ is graphed below. On the same set of axes below sketch $y = g(x-2) + 3$ 2



16. The graph of the function $f(x) = x^2$ is translated m units to the left, dilated vertically by a scale factor of k and then translated 7 units up. The equation of the transformed function is $g(x) = \frac{x^2}{3} + 2x + 10$. Find m and k .

2

$$\begin{aligned} g(x) &= \frac{1}{3}(x^2 + 6x + 30) \\ &= \frac{1}{3}(x^2 + 6x + 3^2 + 30 - 3^2) \\ &= \frac{1}{3}[(x+3)^2 + 21] \\ &= \frac{1}{3}(x+3)^2 + 7 \end{aligned} \quad \begin{aligned} m &= 3 \\ k &= \frac{1}{3} \end{aligned}$$

17. Consider the function $f(x) = (x+1)^2$.

4

The following transformations were applied to the graph of $y = f(x)$ in the order shown below.

- Horizontal translation to the right by 4 units
- Horizontal dilation by a factor of $\frac{4}{3}$
- Vertical dilation by a factor of $\frac{1}{3}$
- Vertical translation down by 3 units
- Reflection in the x -axis

Find the equation of the transformed function and sketch its graph, showing the location of the vertex and the axis intercepts.

$f(x) = (x+1)^2$, which is a concave up parabola with vertex at $(-1, 0)$.

Consider the horizontal transformations:

- Horizontal translation to the right by 4 units:
 - $f(x) = (x+1-4)^2 \rightarrow f(x) = (x-3)^2$
- Horizontal dilation by a factor of $\frac{4}{3}$:
 - $f(x) = \left(\frac{3x}{4} - 3\right)^2$

Consider the vertical transformations:

- Vertical dilation by a factor of $\frac{1}{3}$:
 - $f(x) = \frac{1}{3}\left(\frac{3x}{4} - 3\right)^2$
- Vertical translation down by 3 units:
 - $f(x) = \frac{1}{3}\left(\frac{3x}{4} - 3\right)^2 - 3$
- Reflection in the x -axis:
 - $f(x) = -\frac{1}{3}\left(\frac{3x}{4} - 3\right)^2 + 3$

Which is a concave down parabola with vertex at $(4, 3)$, passing through the origin with the other x -intercept at $x = 8$.

- $$y = (x - 3 - 7)^3 - 6 - 5$$
- $$y = (x - 10)^3 - 11$$

-

$$C = 2$$

$$Pt(4,1)$$

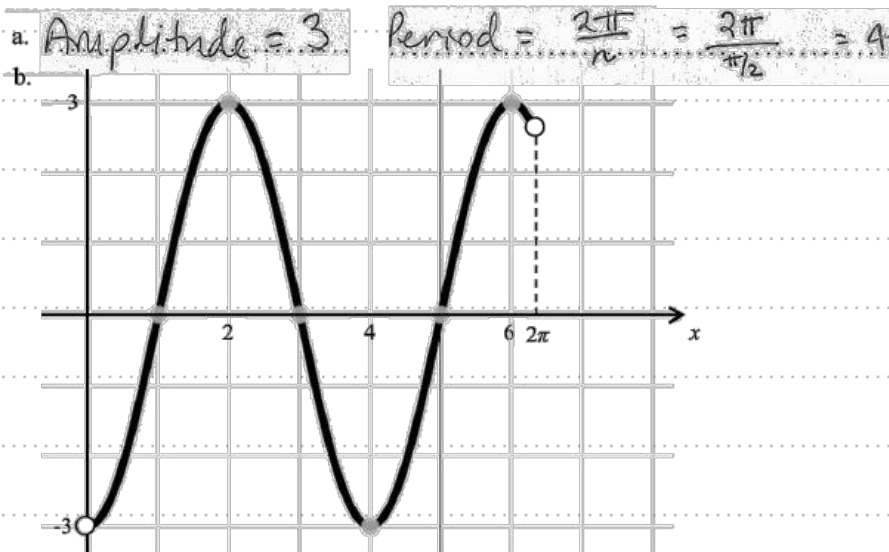
$$1 = \frac{a}{4-3} + 2 \quad \checkmark$$

$$1 \leq a \leq 2$$

21. For the function $f(x) = -3\cos\left(\frac{\pi x}{2}\right)$,

a. Find the amplitude and period. 2

b. Sketch the curve over the domain $\{0 < x < 2\pi\}$. 3



22. Which of the following represents a function $f(x)$ whose graph has undergone the transformation in the following order? 2

- Translated left 2 units
- Horizontally dilated by a factor of 3
- Translated down 4 units

$$f\left(\frac{x+2}{3}\right) - 4$$

23. Describe three transformations which, when applied in succession, change the graph of $y = x^2 + 5$ to the graph with equation $y = \left(\frac{x+1}{2}\right)^2$. 3

$$x^2 + 5 \rightarrow x^2 \rightarrow \left(\frac{x}{2}\right)^2 \rightarrow \left(\frac{x+1}{2}\right)^2$$

Translation 5 units down

Horizontal dilation scale factor 2

Translation 1 unit left